1. Introduction

The theory of Schubert polynomials, initiated by Lascoux and Schützenberger, provides polynomial representatives for Schubert classes in the cohomology of flag varieties. In the classical setting, the Littlewood–Richardson coefficients $c_{u,v}^{w}$ appearing in

$$\mathfrak{S}_u(\mathbf{x}) \cdot \mathfrak{S}_v(\mathbf{x}) = \sum_{w} c_{u,v}^{w}\, \mathfrak{S}_w(\mathbf{x})$$

are nonnegative integers, reflecting the geometric fact that they count intersection points of Schubert varieties. Graham's seminal theorem [**?**] established that in the equivariant setting these coefficients become polynomials in the negative roots x_i with nonnegative integer coefficients.

The quantum deformation of this story was developed by Kirillov and Maeno [**?**], who introduced quantum double Schubert polynomials $\widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{y})$. In the quantum setting the product involves formal variables q^{β} encoding curve-counting information:

$$\widetilde{\mathfrak{S}}_u(\mathbf{x};\mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x};\mathbf{z}) = \sum_{d}\sum_{w} q^{d}\, c_{u,v}^{w,(d)}(\mathbf{y},\mathbf{z})\, \widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{z}). \tag{1}$$

Mihalcea [**?**] proved that when $\mathbf{y} = \mathbf{z}$ (the double case), the coefficients $c_{u,v}^{w,(d)}(\mathbf{y},\mathbf{y})$ are polynomials in the negative roots $y_i - y_j$ with nonnegative integer coefficients, establishing the Peterson conjecture for equivariant quantum cohomology.

In this paper we prove the corresponding positivity for the *triple* case, where $\mathbf{y} \neq \mathbf{z}$ are independent variable sets.

The proof strategy follows the method of Mihalcea, combined with the triple Schubert calculus established by Gao and Xiong in [**?**]. The key idea is a reduction from the quantum triple case to the already-established double positivity of Mihalcea, via a specialization argument using the separated descents condition of [**?**].

2. Notations and Preliminaries

2.1. **Flag variety and Weyl group.** Let $G = SL_n(\mathbb{C})$ and $B \subset G$ a Borel subgroup of upper triangular matrices. The **flag variety** is $Fl_n = G/B$, a smooth complex algebraic variety of dimension $\binom{n}{2}$. Its Weyl group is the symmetric group S_n, generated by simple reflections s_i $(i = 1, \dots, n-1)$ satisfying the Coxeter relations. For $w \in S_n$ the **length** $\ell(w)$ equals the number of inversions of w, and $w_0 = (n, n-1, \dots, 1)$ denotes the longest element.

The set of **descents** of w is

$$\mathrm{Des}(w) = \{\, i : w(i) > w(i+1) \,\}.$$

We say (u, v) satisfies the **separated descents condition** if $\max \mathrm{Des}(u) \le \min \mathrm{Des}(v)$; see [**?** , Definition 2.1].

2.2. **Divided difference operators.** For $i = 1, \dots, n-1$, the **divided difference operator** ∂_i acts on polynomials in variables $x_1, \dots, x_n$ by

$$\partial_i f = \frac{f(\dots, x_i, x_{i+1}, \dots) - f(\dots, x_{i+1}, x_i, \dots)}{x_i - x_{i+1}}.$$

These operators satisfy $\partial_i^2 = 0$ and the braid relations, so for any $w \in S_n$ the operator $\partial_w = \partial_{i_1} \cdots \partial_{i_k}$ is well-defined when $(i_1, \dots, i_k)$ is a reduced word for w.

The **skew divided difference operator** for $v \le w$ in Bruhat order is

$$\partial_{w/v} = \sum_{\substack{u \\ v \le u \le w}} (-1)^{\ell(w)-\ell(u)} \partial_u.$$

2.3. **Schubert polynomials.** The (single) Schubert polynomials of Lascoux–Schützenberger are defined by

$$\mathfrak{S}_w(\mathbf{x}) = \partial_{w^{-1}w_0}(x_1^{n-1}x_2^{n-2}\cdots x_n^0).$$

The **double Schubert polynomials** introduce a second variable set:

$$\mathfrak{S}_w(\mathbf{x};\mathbf{y}) = \partial^{(\mathbf{x})}_{w^{-1}w_0} \prod_{i+j\le n} (x_i + y_j).$$

The **quantum double Schubert polynomials** $\widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{y})$ were introduced in [**?**]. They depend on n variables $\mathbf{x} = (x_1,\ldots,x_n)$, another n variables $\mathbf{y} = (y_1,\ldots,y_n)$, and $n-1$ quantum variables $\mathbf{q} = (q_1,\ldots,q_{n-1})$. The top polynomial is

$$\widetilde{\mathfrak{S}}_{w_0}(\mathbf{x};\mathbf{y}) = \prod_{i=1}^{n-1} \Delta_i(y_{n-i} \mid x_1,\ldots,x_i),$$

where

$$\Delta_k(t \mid x_1,\ldots,x_k) = \sum_{j=0}^{k} t^{k-j} e_j(x_1,\ldots,x_k \mid q_1,\ldots,q_{k-1}),$$

and $e_j(\cdot \mid \mathbf{q})$ are the **quantum elementary symmetric functions**. For any $w \in S_n$,

$$\widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{y}) = \partial^{(\mathbf{y})}_{ww_0} \widetilde{\mathfrak{S}}_{w_0}(\mathbf{x};\mathbf{y}).$$

When $q=0$ we recover the classical double Schubert polynomial: $\widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{y})\big|_{q=0} = \mathfrak{S}_w(\mathbf{x};\mathbf{y})$.

2.4. **Quantum product and structure coefficients.** The **quantum product** in $QH_T^*(Fl_n)$ is defined by the Gromov–Witten correspondence (see [**?** , Definition 1.1] and [**?**]):

$$\sigma(u) \star \sigma(v) = \sum_{d,w} q^d \, c_{u,v}^{w,d} \, \sigma(w),$$

where $c_{u,v}^{w,d}$ are the **equivariant quantum Littlewood–Richardson coefficients**. When $\mathbf{y} = \mathbf{z}$, (2) coincides with the product of Schubert classes in $QH_T^*(Fl_n)$.

The coefficients in (2) satisfy the following specializations:

- When $d=0$, $c_{u,v}^{w,(0)}(\mathbf{y},\mathbf{z})$ equals the triple Schubert coefficient of [**?**].
- When $\mathbf{y} = \mathbf{z}$, they specialize to Mihalcea's coefficients $c_{u,v}^{w,d}$ from [**?**].
- When $\mathbf{y} = \mathbf{z} = 0$ and $d=0$, they give the classical $c_{u,v}^{w}$.

3. Main Conjecture

Consider the product of two quantum double Schubert polynomials with distinct variable sets $\mathbf{y}$ and $\mathbf{z}$:

$$\widetilde{\mathfrak{S}}_u(\mathbf{x};\mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x};\mathbf{z}) = \sum_{d=(d_1,\ldots,d_{n-1})} \sum_{w\in S_n} q_1^{d_1}\cdots q_{n-1}^{d_{n-1}} \, c_{u,v}^{w,(d)}(\mathbf{y},\mathbf{z}) \, \widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{z}). \tag{2}$$

Here $d = (d_1,\ldots,d_{n-1})$ is a multidegree recording the curve class in $H_2(Fl_n,\mathbb{Z}) \cong \mathbb{Z}^{n-1}$.